[2]:
```python
import pandas as pd
data = { 'Name':['Varun','Ragha','Vaibhav'],
         'Age':[25,30,40],
         'Salary':[50000,60000,70000] }
df = pd.DataFrame(data)
print("DataFrame from Scratch:")
print(df)
df['Bonus']=df['Salary']*0.10
print("\nDataFrame after adding 'Bonus' column:")
print(df)
```

```
DataFrame from Scratch:
      Name  Age  Salary
0    Varun   25   50000
1    Ragha   30   60000
2  Vaibhav   40   70000

DataFrame after adding 'Bonus' column:
      Name  Age  Salary   Bonus
0    Varun   25   50000  5000.0
1    Ragha   30   60000  6000.0
2  Vaibhav   40   70000  7000.0
```

[4]:
```python
import pandas as pd
df=pd.read_csv('work experience.csv')
print("First 6 rows of DataFrame")
print(df.head(6))
print("\nColumn names and data types:")
print(df.info())
```

```
First 6 rows of DataFrame
   YearsExperience
0              1.1
1              1.3
2              1.5
3              2.0
4              2.2
5              2.9

Column names and data types:
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 1 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   YearsExperience  9 non-null      float64
dtypes: float64(1)
memory usage: 204.0 bytes
None
```

[13]:
```
import pandas as pd
df=pd.read_csv("values.csv")
print(df)
print("\nMissing values in the DataFrame:")
print(df.isnull().sum())
df.Q1=df['Q1'].fillna(df['Q1'].mean())
print(df.Q1)
```

```
    Year  McDonalds_Revenue_$Billion  Growth_rate_percent   Q1   Q2   Q3   Q4
0   1999                        13.3                  NaN  NaN  NaN  NaN  NaN
1   2000                        14.2                  7.0  NaN  NaN  NaN  NaN
2   2001                        14.9                  4.0  NaN  NaN  NaN  NaN
3   2002                        15.4                  4.0  NaN  NaN  NaN  3.0
4   2003                        17.1                 11.0  3.8  4.3  4.5  4.6
5   2004                        18.6                  8.0  4.4  4.7  4.9  4.5
6   2005                        19.1                  3.0  4.8  5.1  5.3  3.9
7   2006                        20.9                  9.0  4.9  5.4  5.5  5.1
8   2007                        22.8                  9.0  5.3  5.8  5.9  5.8
9   2008                        23.5                  3.0  5.6  6.1  6.3  5.6
10  2009                        22.7                 -3.0  5.1  5.6  6.0  6.0
11  2010                        24.1                  6.0  5.6  5.9  6.3  6.2
12  2011                        27.0                 12.0  6.1  6.9  7.2  6.8
13  2012                        27.6                  2.0  6.5  6.9  7.2  7.0
14  2013                        28.1                  2.0  6.6  7.1  7.3  7.1
15  2014                        27.4                 -2.0  6.7  7.2  7.0  6.6
16  2015                        25.4                 -7.0  6.0  6.5  6.6  6.3
17  2016                        24.6                 -3.0  5.9  6.3  6.4  6.0
18  2017                        22.8                 -7.0  5.7  6.0  5.8  5.3
19  2018                        21.3                 -7.0  5.1  5.4  5.4  5.4
20  2019                        21.4                  1.0  5.0  5.3  5.6  5.4
21  2020                        19.2                -10.0  4.7  3.8  5.4  5.3
22  2021                        23.2                 21.0  5.1  5.9  6.2  6.0
23  2022                        23.2                  0.0  5.7  5.7  5.9  5.9

Missing values in the DataFrame:
Year                          0
McDonalds_Revenue_$Billion    0
Growth_rate_percent           1
```

```
21  2020                        19.2                -10.0  4.7  3.8  5.4  5.3
22  2021                        23.2                 21.0  5.1  5.9  6.2  6.0
23  2022                        23.2                  0.0  5.7  5.7  5.9  5.9

Missing values in the DataFrame:
Year                          0
McDonalds_Revenue_$Billion    0
Growth_rate_percent           1
Q1                            4
Q2                            4
Q3                            4
Q4                            3
dtype: int64
0     5.43
1     5.43
2     5.43
3     5.43
4     3.80
5     4.40
6     4.80
7     4.90
8     5.30
9     5.60
10    5.10
11    5.60
12    6.10
13    6.50
14    6.60
15    6.70
16    6.00
17    5.90
18    5.70
19    5.10
20    5.00
21    4.70
22    5.10
23    5.70
Name: Q1, dtype: float64
```